

# Tolerance Analysis Report

## Project: Bearing Housing Axial Stack-Up

Generation Date: 2026-07-22 12:43  
Standard Mode: ASME Y14.5-2018  
Unit System: Inches

## Tolerance Chain

Chain Name: Axial Preload Shim Gap  
Description: Stack-up from housing face to snap ring groove. The resulting gap dimension represents the spa  
Contributors: 5

## Contributor Data

| # | Name                      | Nominal | Upper Tol | Lower Tol | Direction | Type      |
|---|---------------------------|---------|-----------|-----------|-----------|-----------|
| 1 | Housing Bore Depth        | 1.5000  | ±0.0020   | ±0.0020   | +         | Bilateral |
| 2 | Bearing Width             | 0.7500  | ±0.0005   | ±0.0005   | +         | Bilateral |
| 3 | Spacer Ring               | 0.1250  | ±0.0010   | ±0.0010   | +         | Bilateral |
| 4 | Snap Ring Groove Position | 2.2860  | ±0.0030   | ±0.0030   | –         | Bilateral |
| 5 | Snap Ring Thickness       | 0.0420  | ±0.0020   | ±0.0020   | –         | Bilateral |

## Monte Carlo Simulation (ASME Y14.5)

Iterations: 10,000

| Statistic                | Value  |
|--------------------------|--------|
| Mean                     | 0.0470 |
| Standard Deviation       | 0.0016 |
| Minimum                  | 0.0413 |
| Maximum                  | 0.0525 |
| Percentile 0.135% (–3σ)  | 0.0423 |
| Percentile 2.275% (–2σ)  | 0.0439 |
| Percentile 50% (Median)  | 0.0470 |
| Percentile 97.725% (+2σ) | 0.0501 |
| Percentile 99.865% (+3σ) | 0.0516 |

## **Notes & Assumptions**

- Rule #1 (Envelope Principle) applies unless otherwise specified
- All tolerances converted to bilateral form for analysis.
- RSS analysis assumes independent, normally distributed manufacturing variation.